MHA (without KV-caching)

- **Dimensions:**

$X \in \mathbb{R}^{(seq,\, dm)}$

$W_Q \in \mathbb{R}^{(dm,\, dm)}$

$W_K \in \mathbb{R}^{(dm,\, dm)}$

$W_V \in \mathbb{R}^{(dm,\, dm)}$

seq = length of sequence input

dm = embeding dimension

H = nr. of heads. (divide embeding!)

- **Calculate Q, K, V**

$$Q = X \cdot W_Q \ , \ K = X \cdot W_K \ , \ V = X \cdot W_V$$

$$Q, K, V \in \mathbb{R}^{(seq,\, dm)}$$

$$\to Q_i, K_i, V_i \in \mathbb{R}^{(seq,\, dm/H)}$$

- **Computational Cost for Projections**

$$X^{(seq,\, dm)} \cdot W_Q^{(seq,\, dm)} = \boxed{O(seq \cdot dm^2)}$$

$(seq, dm) \cdot (seq, dm)$
$= O(seq \cdot seq \cdot dm)$

same for W_K, W_V

$$\text{Computation\_total} = \boxed{O(3 \cdot seq \cdot dm^2)} \quad C_1$$

- **Memory Cost:** $\boxed{O(3 \cdot seq \cdot dm)}$, store Q, K, V in memory.

$$\underbrace{(seq, dm)}_{Q} + \underbrace{(seq, dm)}_{K} + \underbrace{(seq, dm)}_{V}$$

- Scaled Dot-Product Attention

$$\text{Attention}_i = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) \cdot V_i$$

Dimension :

- $Q_i, K_i, V_i \in \mathbb{R}^{(seq,\ d_m/H)}$
- $K_i^T \in \mathbb{R}^{(d_m/H,\ seq)}$
- $(Q_i \cdot K_i^T) \in \mathbb{R}^{(seq,\ seq)}$

Computational Cost : for $\boxed{QK^T}$ per head

$$\left(seq, \frac{d_m}{H}\right) \cdot \left(\frac{d_m}{H}, seq\right) = O\left(seq \cdot \frac{d_m}{H} \cdot seq\right)$$
$$= \boxed{O\left(seq^2 \cdot \frac{d_m}{H}\right)}$$

for H heads
$$= O\left(\cancel{H} \cdot seq^2 \cdot \frac{d_m}{\cancel{H}}\right)$$
$$= \boxed{O(seq^2 \cdot d_m)}$$

Memory Cost for Attention

- Dimension : $\mathbb{R}^{(seq,\ seq)}$ per head. $(Q_i \cdot K_i^T)$
- Total H heads : $\boxed{O(H \cdot seq^2)}$

pp3 softmax and V

$$Att_i \times V_i \in \mathbb{R}^{(n_2,\, d_m/H)}$$

- Dimensions:
  - $Att_i \in \mathbb{R}^{(n_2,\, n_2)}$
  - $V_i \in \mathbb{R}^{(n_2,\, d_m/H)}$

- Cost $Att_i \times V_i$ per head

$$(n_2, n_2) \cdot (n_2, \tfrac{d_m}{H}) = O(n_2 \cdot n_2 \cdot \tfrac{d_m}{H}) = O(n_2^2 \cdot \tfrac{d_m}{H})$$

→ for H heads:

$$\underline{Cost} = O(H \cdot n_2^2 \cdot \tfrac{d_m}{H}) = \boxed{O(n_2^2 \cdot d_m)}$$

- Memory Cost $= \boxed{O(n_2 \cdot d_m)}$ for attention output dim in $A \in \mathbb{R}^{(n_2,\, d_m)}$!

- Concat and final Projection

$$MHA_{output} = concat(Att_1, Att_2 \ldots Att_H)$$

dim_final: $MHA_{output} \in \mathbb{R}^{(n_2,\, d_m)}$

Final Projection: $Output = MHA_{output}\, W^O$

where $W_O \in \mathbb{R}^{(d_m,\, d_m)}$

Computational_Cost $= \boxed{O(n_2 \cdot d_m^2)}$

Computational Cost

① Linear projections Q, K, V

$\boxed{O(3 \cdot seq \cdot dm^2)}$

② Att score, calculus QK^T for all heads

$\boxed{O(seq^2 \cdot dm)}$

③ Apply softmax and $Att_i \cdot V$

$\boxed{O(seq^2 \cdot dm)}$

④ Final Projection Wo

$\boxed{O(seq \cdot dm^2)}$

Total Cost = $\boxed{O(4 \cdot seq \cdot dm^2 + 2 \cdot seq^2 \cdot dm)}$

~~Computat~~ Memory Cost

① Memory Q, K, V (all sequence + all heads)

$\boxed{O(3 \cdot seq \cdot dm)}$

② Memory for attention matrix $Q \cdot K^T$

$\boxed{O(H \cdot seq^2)}$

③ Memory $Att_i \cdot V$

$\boxed{O(seq \cdot dm)}$

Total Memory =

$\boxed{O(3 \cdot seq \cdot dm + H \cdot seq^2)}$